

2 The photograph shows a thistle plant.



(Source: © John Stacey)

This plant grows in grassland and is not eaten by cattle.

A student investigated the effect of treating thistle plants with a solution of insecticide. Other thistle plants were treated with water.

All the thistle plants used were selected, at random, from the same field.

After 56 days, the student measured the longest leaf of each thistle plant used in this investigation.

(a) State a suitable null hypothesis for this investigation.

(1)

(b) The table shows the results of this investigation.

Insecticide leaf length / cm	Water leaf length / cm
12.6 13.1 13.9 14.0 11.1	9.1 9.4 10.3 10.1 10.4 10.9
11.6 12.4 12.9 12.5 13.4	9.0 9.7 9.9 10.8 9.3 9.8
13.7 12.0 11.9 11.8 12.6	9.2 9.7 9.4

- (i) Calculate the percentage difference between the shortest and longest leaves treated with insecticide.

(1)

Answer%

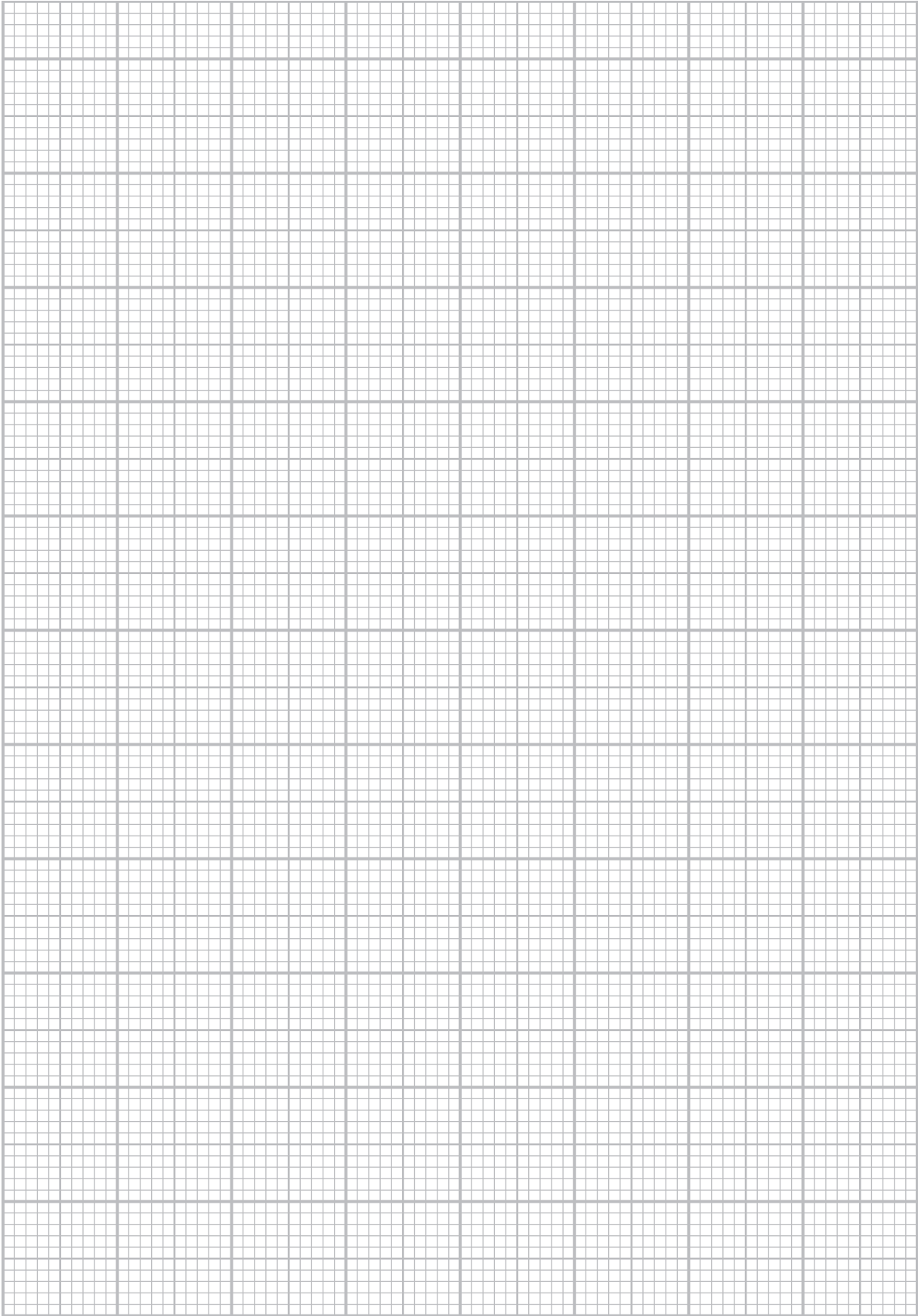
- (ii) Calculate the mean length of the leaves for plants treated with insecticide and the mean length of the leaves for plants treated with water.

Draw a suitable table to display the **raw data** and your calculated **mean** for each treatment.

(3)

(c) Plot a suitable graph to compare the results of this investigation.

(3)



(d) (i) The student analysed the data with a t test, using the formula

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{(S_1)^2}{n_1} + \frac{(S_2)^2}{n_2}}}$$

where:

$\bar{x}$ is the mean value for each treatment

n is the number of thistle plants in each treatment

$(S_1)^2 = 0.750$ and $(S_2)^2 = 0.357$

Calculate the value of t .

(3)

Answer

(ii) The table shows the critical values of t for different degrees of freedom.

The number of degrees of freedom = $(n_1 - 1) + (n_2 - 1)$

Degrees of freedom	$P = 0.05$	$P = 0.01$
15	2.131	2.947
16	2.120	2.921
17	2.110	2.898
18	2.101	2.878
19	2.093	2.861
20	2.086	2.845
21	2.080	2.831
22	2.074	2.819
23	2.069	2.807
24	2.064	2.797
25	2.060	2.787
26	2.056	2.779
27	2.052	2.771
28	2.048	2.763
29	2.045	2.756
30	2.042	2.750

Deduce the conclusion that can be drawn from this investigation.

Use your graph and the information in the table to support your answer.

(2)

[illegible]

(e) Explain **two** ways in which this investigation could be improved.

(2)

1.....

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2.....

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(Total for Question 2 = 15 marks)
